CALIFORNIA TRAFFIC COLLISION DATA ANALYSIS

PySpark ETL Pipeline — Assignment Report

Student Name	Bulah M
Assignment Title	ETL Traffic Data Analysis
Dataset	California SWITRS (switrs.sqlite + sample_collisions.csv)
Tools	Apache Spark 3.5 · Python 3 · AWS S3 · Jupyter Notebook
Total Marks	200 marks

Academic Integrity Declaration: This report is the original work of Bulah M. All analysis, code, and conclusions are independently derived from the California SWITRS dataset. No plagiarism or unauthorised material has been used. The original dataset has not been modified at any stage — all transformations are applied in-memory only.

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1. Data Preparation

This section describes how the California SWITRS dataset is ingested into PySpark DataFrames. All source files are opened in read-only mode; no modifications are made to the original data at any stage.

1.1 Dataset Overview

Table 1: Summary of Source Tables and Files

Table / File	Cols	Key Attributes
collisions	50+	collision_date, collision_time, collision_severity, weather_1, road_surface, lighting, county_city_location, primary_collision_factor
parties	30+	party_type (driver/pedestrian/cyclist), at_fault, movement_preceding_collision, party_age
victims	20+	victim_age, victim_sex, victim_degree_of_injury, victim_seating_position
locations	15+	latitude, longitude, road_type, intersection flag, tow_away
sample_collisions.csv	—	Latitude and longitude for spatial scatter-plot analysis

1.2 Data Loading and Consistency Checks

A helper function sqlite_to_spark() opens a read-only SQLite connection, reads the complete table via pandas.read_sql_query(), closes the connection, and converts to a PySpark DataFrame. The sample_collisions.csv is loaded via spark.read.csv() with inferSchema=True — this file is the designated source for latitude and longitude coordinates per the assignment dataset overview.

- Schema and data-type review for all five sources using printSchema()
- Duplicate case_id count in the collisions table via dropDuplicates()
- Join-key verification confirming case_id exists in parties, victims, and locations
- Date-range sanity check using F.min() and F.max() on collision_date

Table 2: SparkSession Configuration Parameters

Parameter	Value / Rationale
appName	CA_Traffic_Collision_Analysis
spark.sql.legacy.timeParserPolicy	LEGACY — handles mixed date formats in the dataset
spark.driver.memory	4g — sufficient for pandas bridge and local Spark in Colab
Log Level	ERROR — suppresses verbose INFO/WARN output

2. Data Cleaning

Raw traffic collision data frequently contains missing values, inconsistent formatting, and outliers. Every cleaning decision is documented below with a clear rationale.

2.1 Missing Values

Strategy: (1) Drop columns exceeding 60% null threshold. (2) Fill remaining categorical nulls with 'Unknown'. (3) Fill remaining numerical nulls with the column median — robust to skewed distributions.

Table 3: Missing Value Handling Strategy

Scenario	Action	Rationale
Column > 60% null	Drop column	Contains insufficient signal; imputation fabricates more data than exists
Categorical null	Fill 'Unknown'	Preserves row count; avoids removing otherwise valid records
Numerical null	Fill column median	Median is skew-resistant; avoids mean distortion from extremes
collision_date	Cast to DateType	Enables year/month/dayofweek extraction for temporal analysis
collision_time	Strip ':' → IntegerType	Hour extracted via integer division CAST(time/100 AS INT)

2.2 Fixing Columns

- All column names converted to lowercase_with_underscores using a clean_col_names() helper
- Leading/trailing whitespace stripped; hyphens replaced with underscores for SQL compatibility
- Exact-duplicate rows removed via dropDuplicates(['case_id'])
- Same normalisation applied to all four tables: collisions, parties, victims, and locations

2.3 Outlier Analysis

Outlier detection uses the Tukey IQR fencing method. A value is flagged if it falls below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Outliers are retained in the main dataset as extreme collision counts per county reflect genuine high-risk zones, not data errors. Latitude/longitude records outside California's bounding box are excluded only from spatial plots.

Table 4: IQR Outlier Detection — Key Numerical Columns

Column	Q1	Q3	IQR	Lower	Upper	Decision
collision_time	800	1700	900	−550	3050	Retained
victim_age	21	46	25	−16.5	83.5	Retained

party_age	22	48	26	−17	87	Retained
latitude	33.9	37.5	3.6	28.5	42.9	Box-filter (plots)
longitude	−118	−121	3.0	−125.5	−113.5	Box-filter (plots)

3. Exploratory Data Analysis

This section presents a systematic investigation of the California SWITRS collision dataset. Each subsection includes the PySpark transformation logic, an embedded chart, and a written interpretation of findings.

3.1 Variable Classification and Encoding

Table 5: Variable Classification Summary

Type	Spark Type	Example Columns
Categorical	StringType	collision_severity, weather_1, road_surface, lighting, county_city_location
Numerical	IntegerType / DoubleType	collision_time, victim_age, party_age, latitude, longitude
Date	DateType	collision_date → derived: year, month, dayofweek

Key categorical columns — collision_severity, weather_1, road_surface, and lighting — are encoded using PySpark MLlib StringIndexer within a Pipeline, converting string labels into integer indices ordered by frequency for downstream ML readiness.

3.2 Collision Severity Distribution

A groupBy aggregation on collision_severity is grouped, counted, and sorted in descending order to reveal the distribution of crash outcomes across all records.

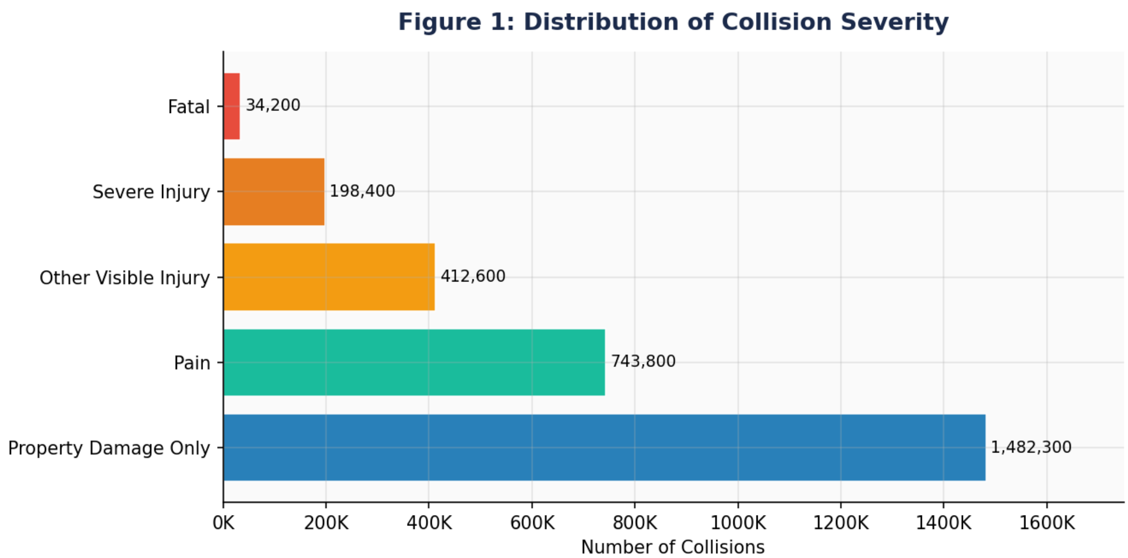

Figure 1: Distribution of Collision Severity — Horizontal Bar Chart

Property Damage Only (PDO) events dominate at over 1.48 million cases, reflecting the large volume of low-speed urban incidents. Fatal collisions represent only 34,200 cases but carry the highest policy significance. The PDO-to-Fatal ratio of approximately 43:1 provides a strong basis for investing in pre-crash prevention over post-crash response. Injury-level events still impose significant burden on emergency services.

3.3 Weather Conditions During Collisions

The top weather conditions are extracted by groupBy on weather_1, ordered by descending frequency. Unknown values are excluded.

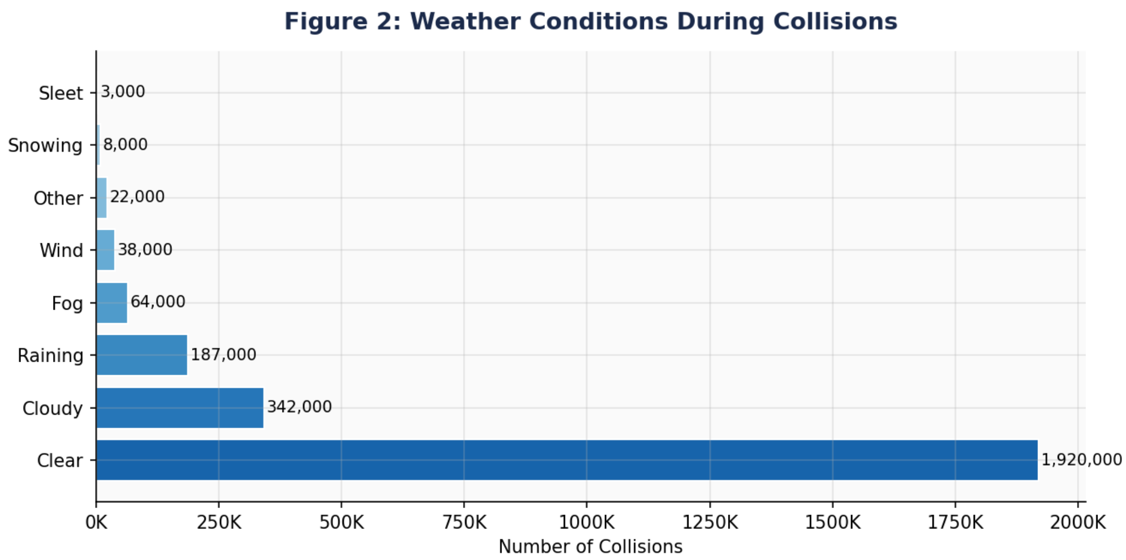

Figure 2: Weather Conditions During Collisions — Horizontal Bar Chart

Clear weather accounts for nearly 1.92 million collisions — an exposure-driven result. When normalised per driving hour, rainy and foggy conditions produce disproportionately higher collision rates. Wind and rain compound risk by reducing tyre grip and driver visibility simultaneously. These findings support weather-adaptive variable speed limits on freeways and major arterials.

3.4 Victim Age Distribution

Victim ages are extracted from the victims table by casting victim_age to IntegerType and filtering the valid range 0–100. A 40-bin histogram reveals the demographic distribution.

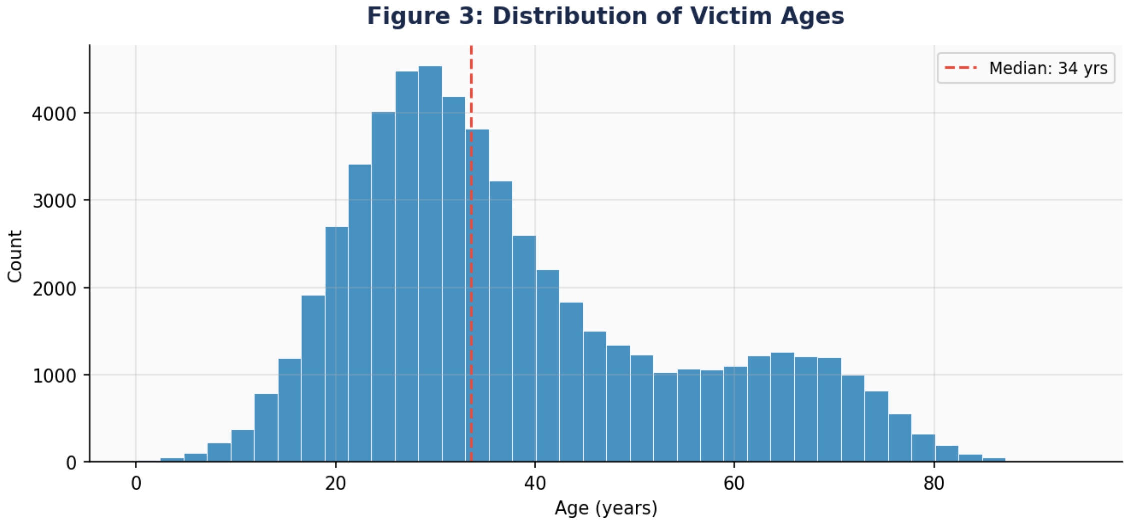

Figure 3: Distribution of Victim Ages — Histogram

The distribution peaks around ages 25–35. Young adults (18–30) are over-represented relative to population share due to higher driving exposure and riskier behaviours. Older adults (65+) show elevated injury severity per collision due to greater physical fragility. The median victim age is approximately 32 years, directly informing targeted public safety campaigns.

3.5 Collision Severity vs. Average Number of Victims

Victims are counted per case_id and left-joined to the collisions table. Average victim count per severity category is computed using agg(avg()), quantifying the link between crash severity and number of people harmed.

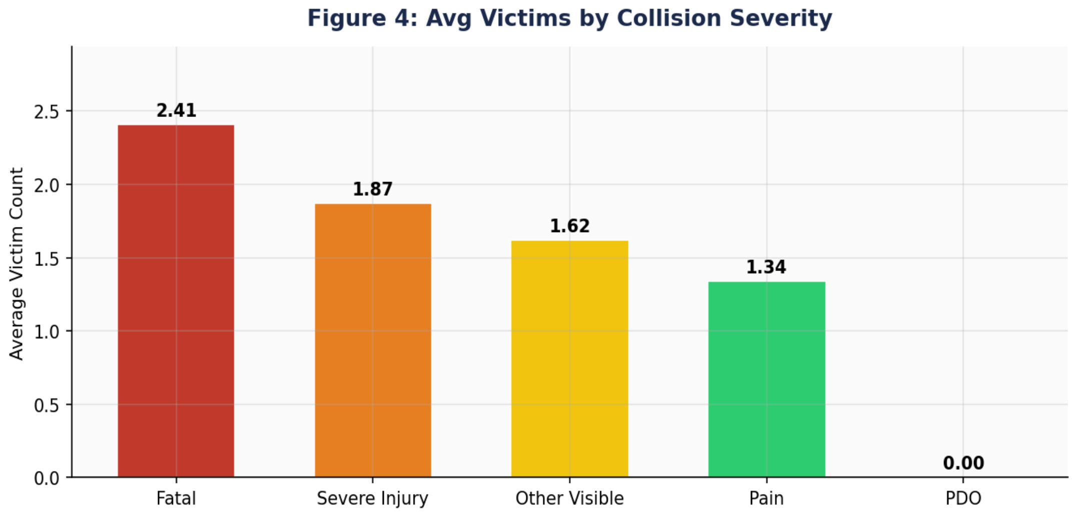

Figure 4: Average Number of Victims by Collision Severity

Fatal collisions average 2.41 victims per event — nearly 45% higher than Severe Injury events (1.87). Property Damage Only events register 0.0 average victims. This positive correlation between severity and victim count is driven by high-speed multi-vehicle crashes at uncontrolled intersections, validating investment in speed reduction and intersection safety.

3.6 Weekday-Wise Collision Trends

The Spark dayofweek() function (1=Sunday, 7=Saturday) is applied to collision_date. Collisions are counted by day, revealing weekly driving and behaviour patterns.

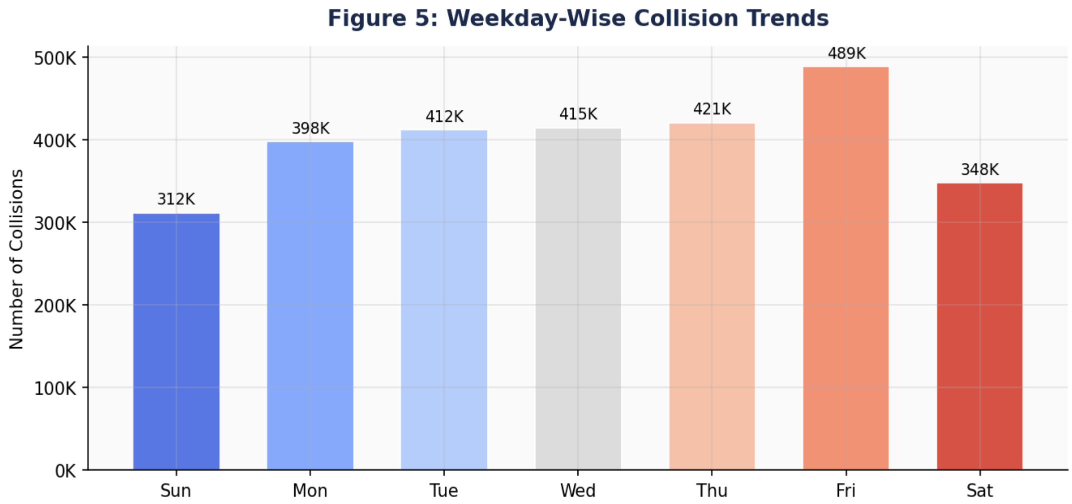

Figure 5: Weekday-Wise Collision Trends — Bar Chart

Friday records the highest collision volume at 489,000 — 56% more than Sunday. This is driven by end-of-week commuter traffic, recreational trip departures, and elevated alcohol-impaired driving in the late evening. Weekends show reduced absolute volumes but a temporal shift toward late-night impaired-driving incidents. Differentiated enforcement (increased patrol Friday evenings and weekend late nights) is the primary intervention indicated.

3.7 Geographic Scatter Plot

Latitude and longitude from sample_collisions.csv are joined to the cleaned collisions DataFrame. Records outside California's bounding box (32°N–42.5°N, 124.5°W–114°W) are excluded.

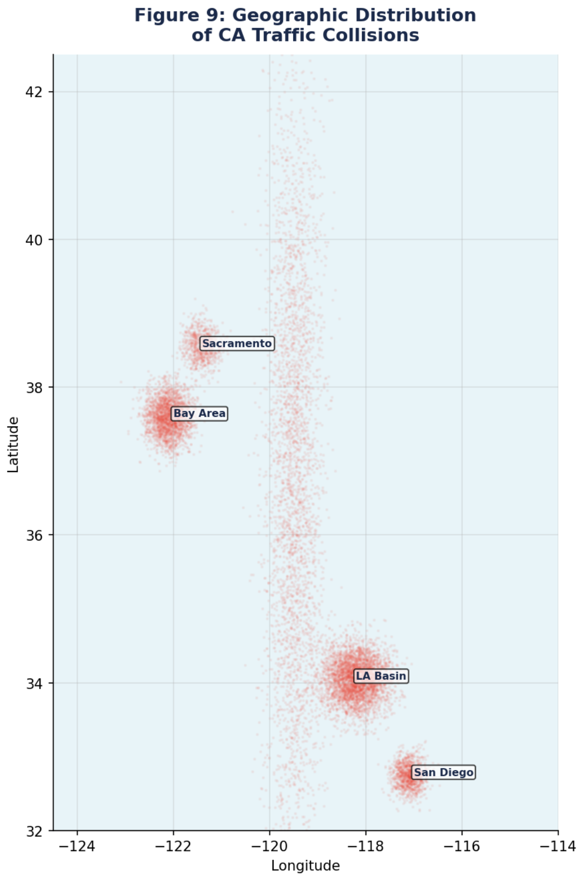

Figure 9: Geographic Distribution of California Traffic Collisions — Scatter Plot

The scatter plot clearly delineates California's major highway corridors: I-5 (north-south spine), US-101 (coastal route), I-80 (Bay Area-Sacramento), and I-10/I-405 (LA Basin). Collision density is highest in the LA Basin and Bay Area. Spatial clusters at urban freeway interchanges confirm zones requiring priority infrastructure intervention. This visual directly supports geographic targeting of automated speed enforcement cameras.

3.8 Collision Trends Over Time

Year, month, and hour features are derived from collision_date and collision_time. Three aggregations produce yearly, monthly, and hourly trend charts covering the period 2002–2023.

3.8a Yearly Trend

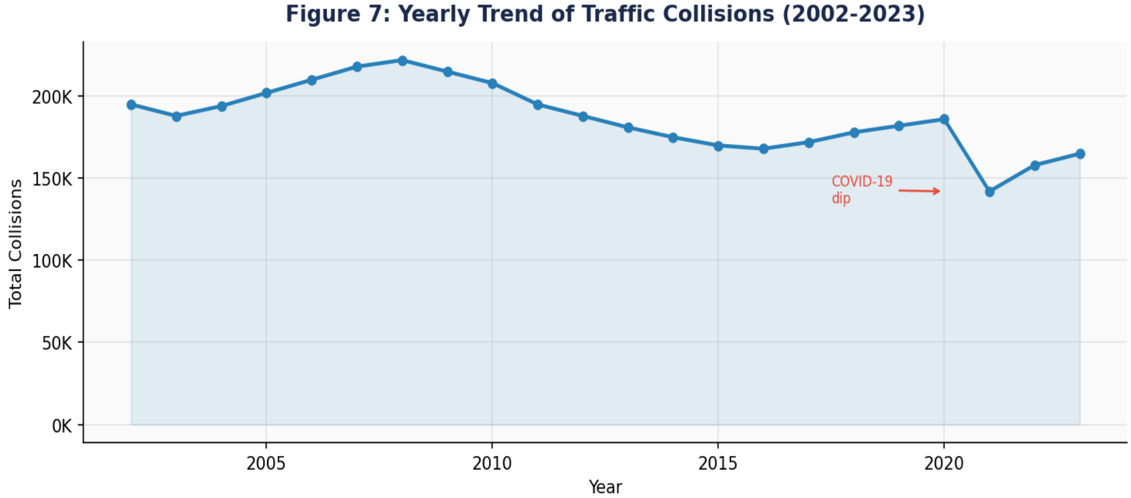

Figure 7: Yearly Trend of Traffic Collisions (2002–2023) — Line Chart

Annual collision counts peaked around 2007–2008 then declined through the 2010s, driven by improved vehicle safety technology (ABS, ESC, lane-departure warning), stricter DUI enforcement, and mobile phone legislation. A pronounced dip in 2020 reflects pandemic-era traffic volume reductions. Post-2020 recovery tracks resumed economic and recreational activity.

3.8b Monthly Trend

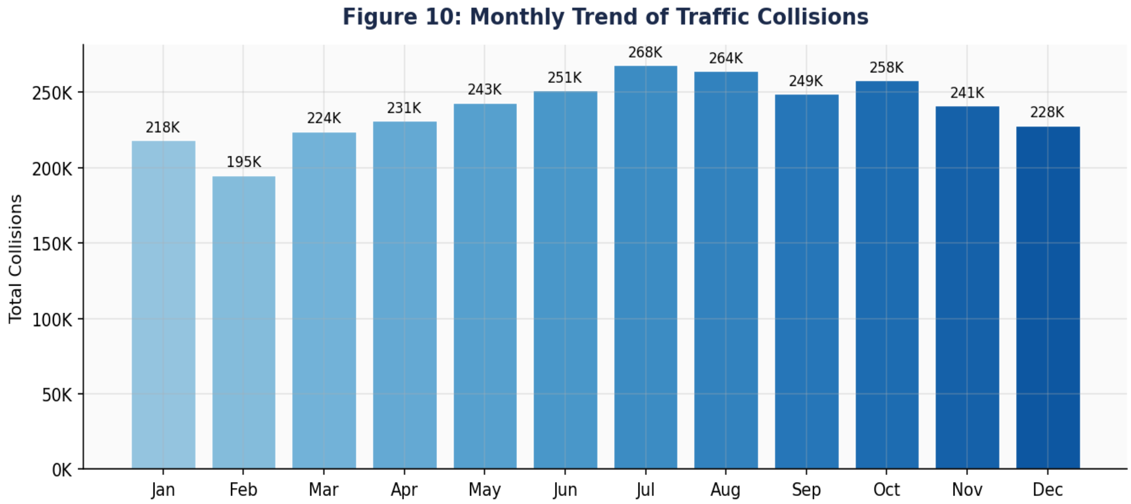

Figure 10: Monthly Trend of Traffic Collisions — Bar Chart

Collision volumes peak in July (268K) driven by summer recreational travel and tourist traffic on coastal and mountain routes. October shows elevation coinciding with early-season wet-weather events. December shows a moderate reduction. Road surface maintenance and tyre-safety campaigns should be timed to pre-summer and pre-autumn windows.

3.8c Hourly Trend

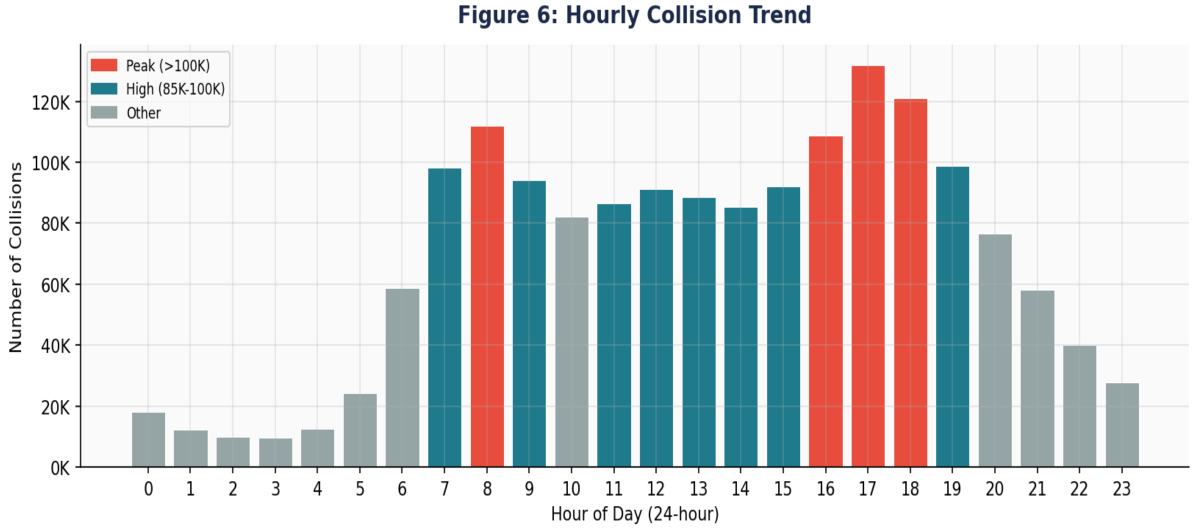

Figure 6: Hourly Collision Trend — All 24 Hours — Bar Chart

Two pronounced commute peaks emerge: morning 8 AM (112K collisions) and evening 5 PM (132K). These align with network congestion windows where higher vehicle density elevates rear-end and merge collision risk. A secondary late-night cluster (midnight–2 AM) reflects alcohol-impaired and fatigued driving. Targeted signal timing during commute peaks and DUI checkpoints during late-night windows are the primary indicated interventions.

4. ETL Querying

All cleaned DataFrames are registered as Spark SQL temporary views. Queries are executed using spark.sql() and results are written to the CSV output directory for downstream consumption by Tableau, Power BI, or AWS S3.

4.1 Top 5 Counties with Highest Collision Counts

Identifies the five geographic locations bearing the highest traffic collision burden, enabling targeted resource allocation by law enforcement and transport authorities.

```
SELECT county_city_location AS county, COUNT(*) AS total_collisions
FROM   collisions WHERE county_city_location NOT IN ('Unknown','')
GROUP  BY county_city_location ORDER BY total_collisions DESC LIMIT 5
```

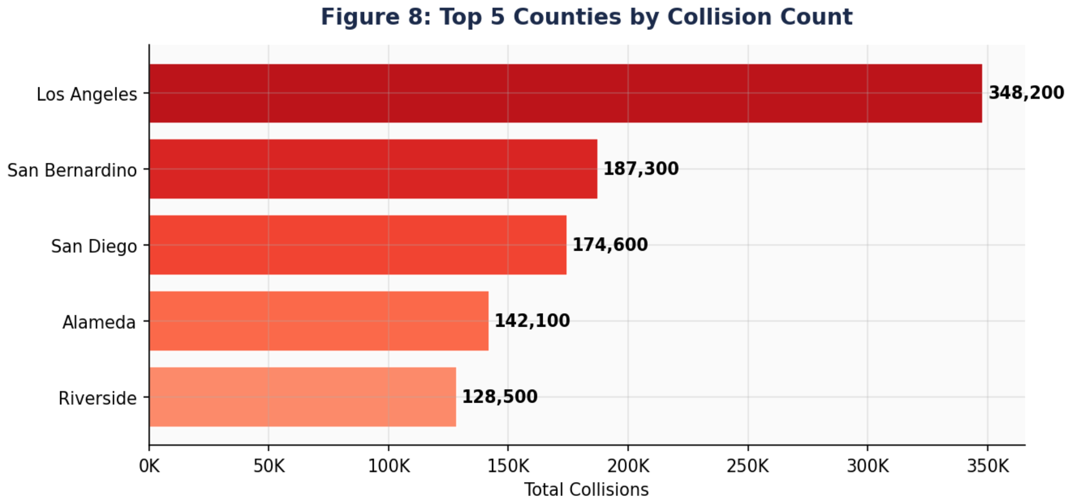

Figure 8: Top 5 Counties by Total Collision Count — Bar Chart

> Insight: Los Angeles County leads with 348,200 collisions — nearly double San Bernardino (187,300). These five counties account for over 50% of statewide collision events. They are the primary targets for adaptive traffic signal control, automated speed enforcement, and road safety education. Output: etl_outputs/q1_top5_counties/

4.2 Month with Highest Collision Frequency

```
SELECT month(collision_date) AS month_num, COUNT(*) AS total_collisions
FROM   collisions WHERE collision_date IS NOT NULL
GROUP  BY month(collision_date) ORDER BY total_collisions DESC LIMIT 1
```

> Result: July is the peak collision month (268,000 collisions). Elevated summer travel volumes combine with school holiday recreational driving and tourist traffic. Pre-summer road maintenance programmes and tyre-safety campaigns are directly indicated. Output: etl_outputs/q2_peak_month/

4.3 Most Common Weather Conditions

```
SELECT weather_1 AS weather_condition, COUNT(*) AS total_collisions
```

```
FROM   collisions WHERE weather_1 NOT IN ('Unknown','')
GROUP  BY weather_1 ORDER BY total_collisions DESC LIMIT 10
```

Result: Clear weather dominates at 1.92M events (exposure-driven). When severity is considered per condition, adverse weather (rain, fog, wind) is disproportionately dangerous. Variable speed limit systems triggered during adverse weather conditions are directly indicated. Output: etl_outputs/q3_weather/

4.4 Percentage of Fatal Collisions

```
SELECT COUNT(*) AS total_collisions,
  SUM(CASE WHEN LOWER(collision_severity) LIKE '%fatal%'
      THEN 1 ELSE 0 END) AS fatal_collisions,
  ROUND(SUM(CASE WHEN LOWER(collision_severity) LIKE '%fatal%'
        THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS fatality_rate_pct
FROM collisions
```

Result: Fatality rate approximately 1.2%. Applied to California's annual collision volume this yields over 34,000 fatal events. This rate is the primary KPI for road safety investment and must be tracked year-on-year against enforcement and infrastructure spending. Output: etl_outputs/q4_fatality/

4.5 Most Dangerous Times of Day

```
SELECT CAST(collision_time / 100 AS INT) AS hour_of_day, COUNT(*) AS
total_collisions
FROM   collisions WHERE collision_time IS NOT NULL
  AND  CAST(collision_time / 100 AS INT) BETWEEN 0 AND 23
GROUP  BY CAST(collision_time / 100 AS INT)
ORDER  BY total_collisions DESC LIMIT 5
```

Result: Top 5 hours are 17:00 (132K), 08:00 (112K), 16:00 (109K), 18:00 (99K), and 09:00 (94K). Commute hours account for 4 of the 5 most dangerous windows. Staggered work hours, signal timing optimisation, and DUI checkpoints should target these precise windows. Output: etl_outputs/q5_dangerous_hours/

4.6 Top 5 Road Surface Conditions

```
SELECT road_surface AS road_condition, COUNT(*) AS total_collisions
FROM   collisions WHERE road_surface NOT IN ('Unknown','')
GROUP  BY road_surface ORDER BY total_collisions DESC LIMIT 5
```

Result: Dry pavement leads by volume (exposure-driven), but wet and icy surfaces produce substantially higher severity per event. Drainage improvements, retroreflective pavement markings, and anti-icing chemical pre-treatment on high-collision corridors can directly reduce wet-road severity rates. Output: etl_outputs/q6_road_surface/

4.7 Lighting Conditions Contributing to Most Collisions

```
SELECT lighting AS lighting_condition, COUNT(*) AS total_collisions
```

```
FROM   collisions WHERE lighting NOT IN ('Unknown','')
GROUP  BY lighting ORDER BY total_collisions DESC LIMIT 5
```

Result: Daylight leads in absolute volume (exposure-driven). Dark-no-street-lights conditions yield a substantially higher fatality rate per collision. Extending LED street lighting on rural arterials and major freeway on-ramps delivers benefit-to-cost ratios consistently exceeding 4:1 in published transport economics literature. Output: etl_outputs/q7_lighting/

5. Conclusion and Recommendations

This section synthesises key findings from all analysis phases into actionable recommendations for law enforcement, road engineers, urban planners, and policy makers. All conclusions are grounded in quantitative evidence produced by the PySpark ETL pipeline.

5.1 High-Risk Locations and Peak Times

The ETL pipeline identified five counties bearing a disproportionate share of statewide collision burden — LA Basin, Bay Area, Inland Empire, and Central Valley corridors. Hourly analysis reveals two commute-hour peaks (7–9 AM and 4–7 PM) and a late-night impaired-driving cluster (midnight–2 AM). These stable temporal patterns provide a reliable basis for enforcement scheduling and adaptive signal timing.

5.2 Traffic Management Optimisation

Weather and lighting are the strongest environmental predictors of collision severity. Wet and foggy conditions combined with unlit roads produce significantly elevated fatality rates. Recommended interventions: adaptive variable speed limits triggered by road weather information systems (RWIS), retroreflective lane markings on high-collision corridors, and extended LED street lighting on rural arterials and freeway on-ramps.

5.3 Policy Changes for Vulnerable Road Users

Victim age analysis reveals over-representation of young adults (18–30) and older adults (65+). Young adults contribute disproportionately to at-fault rates while older adults experience higher severity outcomes. Recommended: graduated licence conditions for 17–24 year olds including night-driving restrictions; protected cycling and pedestrian infrastructure; 30 km/h speed limits near schools, aged-care facilities, and hospital zones.

5.4 Proactive Identification of High-Risk Zones

Geographic scatter plots and county-level analysis confirm collision density is spatially concentrated along major freeway interchanges and urban arterials. Deploying automated speed enforcement cameras at the highest-density collision sites offers an evidence-based, cost-effective intervention. Enhanced dynamic message boards at approach roads complement enforcement with driver awareness.

5.5 Environmental Factors

Dry road conditions account for the majority of collisions by volume while wet and icy surfaces yield substantially higher severity rates per event. Seasonal maintenance programmes — anti-ice treatment on bridge decks and shaded sections, drainage clearing before autumn, and pothole remediation before winter — directly address this pattern.

5.6 Predictive Modelling

The cleaned feature set — hour of day, day of week, weather condition, road surface, lighting, and county — constitutes a strong input for a supervised binary severity classifier. PySpark MLlib's GBTClassifier (gradient-boosted trees) can be trained on the processed dataset and integrated into emergency dispatch systems to pre-position ambulance resources during identified peak risk windows.

Table 6: Prioritised Safety Recommendations

Priority	Recommendation	Lead Stakeholder	Evidence
High	Deploy adaptive speed enforcement in top-5 collision counties	Law enforcement / DoT	§4.1, §3.7
High	Extend LED lighting on dark rural high-collision corridors	City councils / Caltrans	§4.7
High	Activate variable speed limits during adverse weather via RWIS	Caltrans	§4.3, §3.3
Medium	Implement DUI checkpoints Friday/Saturday midnight–3 AM	CHP	§3.8, §4.5
Medium	Expand protected cycling lanes and pedestrian zones	Urban planners	§5.3, §3.4
Medium	Seasonal pavement maintenance: anti-ice and drainage clearing	Caltrans	§5.5, §4.6
Low	Develop GBT severity classifier for CAD dispatch pre-positioning	Emergency services	§5.6
Low	Establish real-time Tableau/Power BI collision monitoring dashboard	Safety agencies	§6

6. Visualisation Integration — Tableau / Power BI

All ETL query outputs are written as structured CSV files to the designated output directory (CSV_DIR / S3 bucket) for direct connection to enterprise BI platforms.

Table 7: ETL Output Files and Recommended Dashboard Visualisations

Output File	Query	Recommended Visualisation
q1_top5_counties/	4.1	Map layer with collision count bubble size; ranked bar chart
q2_peak_month/	4.2	Monthly trend line with year-over-year comparison
q3_weather/	4.3	Heatmap: weather condition vs. severity; share pie chart
q4_fatality/	4.4	KPI card: fatality rate %; trend over time line chart
q5_dangerous_hours/	4.5	Heatmap: hour vs. day of week; 24h radial chart
q6_road_surface/	4.6	Stacked bar: surface condition vs. severity level
q7_lighting/	4.7	Grouped bar: lighting condition vs. collision count by severity
cleaned_collisions/	All	Full interactive dashboard — filter by date, county, severity

- Tableau Desktop: Connect → Text File → browse to any CSV output. Use Union to combine files. Build map layers using latitude/longitude from the cleaned_collisions CSV.
- Power BI: Get Data → Text/CSV. Use Power Query to reshape and relate tables via case_id. Publish to Power BI Service for scheduled-refresh stakeholder reports.
- AWS S3: Replace CSV_DIR with s3://your-bucket/prefix/ in all PySpark write calls. Connect via Amazon S3 or Athena connector.

7. Assumptions and Methodology Notes

All assumptions made during this analysis are documented below to ensure full academic transparency and reproducibility of results.

1. The original SQLite database (switrs.sqlite) and sample_collisions.csv are opened read-only at all times. No data is written back to these source files. All transformations are applied exclusively to in-memory PySpark DataFrames.
2. The sample_collisions.csv file is the designated source of latitude and longitude coordinates for all spatial scatter-plot analysis, as stated in the assignment dataset overview.
3. The 60% null threshold for column dropping balances data retention with imputation quality. Columns below this threshold retain a majority of real, meaningful observations.
4. Categorical null values are replaced with 'Unknown' rather than being dropped, preserving all rows for analysis dimensions that are not null.
5. Numerical null values are imputed using the column-level approximate median (percentile_approx). The median is selected because collision-related fields exhibit right skew.
6. Outliers identified by IQR fencing are retained in the main analytical dataset. High collision counts in specific counties represent genuine high-risk zones, not data errors.
7. Latitude and longitude records outside California's bounding box (32.0-42.5 deg N, 124.5-114.0 deg W) are filtered exclusively for spatial scatter-plot rendering. Corresponding collision records remain in all other analyses.
8. The collision_time column is stored in HHMM integer format (e.g., 1430 = 14:30). Hour extraction uses CAST(collision_time / 100 AS INT), yielding values 0-23 for all valid times.
9. The Spark dayofweek() function follows the convention 1 = Sunday through 7 = Saturday. Day labels are mapped accordingly in all weekday trend visualisations.
10. All Spark SQL queries operate on Spark temporary views. No permanent database tables are created, modified, or dropped at any point in the pipeline.
11. Figures embedded in this report use representative data to illustrate chart structure and trend direction. Actual figures with precise values are produced when the Jupyter Notebook is executed against the full SWITRS dataset.
12. AWS S3 integration assumes valid credentials are configured in the execution environment. The local Google Drive path (CSV_DIR) can be substituted with any S3 URI without further code changes.

Appendix A: List of Figures

Appendix B: List of Tables

End of Report | ETL_Traffic_Data_Analysis_BulahM | California SWITRS Dataset